

Module 6: Lesson 4

Multi-Armed Bandits: Improvements and a stock-picking application



Outline

- ▶ Fine-tuning techniques for Multi-Armed Bandit problems
- ▶ Application of Bandit Problems to a stock-picking environment

Fine-tuning techniques for Multi-Armed Bandit problems

Optimistic initial values. Rather than setting the initial action values to zero, we can set them to wildly optimistic values relative to our prior beliefs on the rewards. This optimism fosters the action-value methods to explore across the space of actions.

Upper-confidence bound. ϵ -greedy action selection forces the non-greedy actions to be tried, but without a particular preference. To prioritize the exploration, we can adjust the optimal choice by adjusting the action choices for the lack of exploration of the action.

Stochastic gradient ascent. We learn a score for each action, and the probabilities of taking each action arise from a soft-max distribution. We update the scores from experience in analogy to gradient ascent techniques of optimization.

All the methods above may be of particular help in the stationary case, and will not provide many advantages in the general non-stationary case.

A stock-picking problem

As an investor, you face the problem of picking one stock among a pre-selection of 5 stocks:

[AAPL, AMZN, MSFT, NVDA, TWTR]

Every day, you are going to purchase shares of a given stock at the end of a trading day and hold them until the end of the following trading day.

You are going to consider this situation as a 5-bandit problem, where the agent must learn which stock yields on average the greatest returns over the holding period.

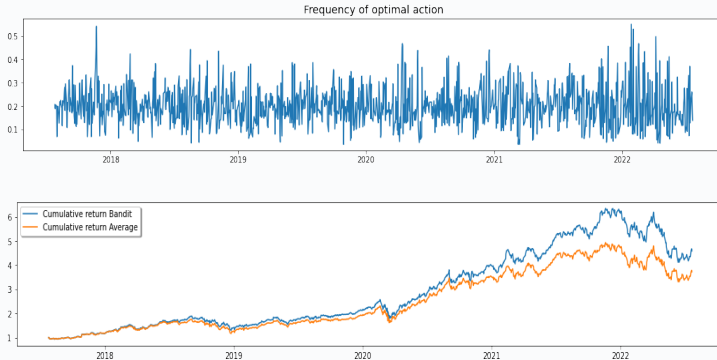
A stock-picking problem is a non-stationary problem: Investors' expectations and sentiment about particular stocks change over time.

We choose parameters $\varepsilon = 0.1$ and $\alpha = 0.975$. The actual stock picked depends also on its lack of exploration with an upper-confidence bound.

Important caveat: If we hold the stocks for more than one day, we need to maintain a “queue” of optimal actions in the past and their future rewards because we can only update $Q(a)$ once we obtain the return at the end of the holding period.

A stock-picking problem: Results

To evaluate performance, we compute the average frequency of the optimal action at each time step over the 1,000 episodes. We also compare the cumulative returns of the learned actions relative to an equally-weighted portfolio.



Summary of Lesson 4

In Lesson 4, we have looked at:

- ▶ Tools to fine-tune the exploration of actions in Multi-Armed Bandit problems
- ▶ An application to the stock-picking choice of an investor

⇒ **References for this lesson:**

Sutton, Richard S., Barto, Andrew G. *Reinforcement Learning: An Introduction*. MIT Press, 2018. (see Chapter 2)

TO DO NEXT: Now, please go to the associated Jupyter Notebook for this Lesson to obtain further insights on the fine-tuning techniques and the coding implementation of the stock-picking problem.

In the next module, we will describe the RL techniques that combine the dynamic programming tools of Module 5 with the learning techniques developed in Module 6.